



Modelling Financial Markets

Chapter 16: Digital Assets and DeFi



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Acronyms used in this material (1/2)

- AI: Artificial Intelligence
- AIC: Akaike Information Criterion
- AR: AutoRegressive (model)
- BET: Bucharest Exchange Trading (index)
- BNB: BNB (native token of BNB Chain, formerly Binance Coin)
- BSC: BNB Smart Chain
- CAPM: Capital Asset Pricing Model
- CI: Confidence Interval
- CMKT: Crypto Market factor (value-weighted market excess return)
- CMOM: Crypto Momentum factor (past winners minus losers)
- COVID-19: COronaVirus Disease 2019
- CRIX: CRypto IndeX
- CSIZE: Crypto Size factor (small minus big coins)
- DAI: Dai (stablecoin of the Maker/Sky protocol)
- DTB4WK: 4-Week Treasury Bill Secondary Market Rate (FRED series code)
- ECRIX: Exact CRIX (number of constituents chosen in steps of one)
- EFCRIX: Exact Full CRIX (global minimum of the AIC)
- ES: Expected Shortfall
- ETF: Exchange-Traded Fund
- ETHA: iShares Ethereum Trust ETF (ticker)
- FDIC: Federal Deposit Insurance Corporation
- FM: Fama–MacBeth (regressions)
- FRED: Federal Reserve Economic Data
- FWER: Family-Wise Error Rate
- GMM: Generalized Method of Moments
- GRS: Gibbons–Ross–Shanken (test)

Acronyms used in this material (2/2)

- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- LP: Liquidity Provider
- LUNA: Luna (native token of the Terra blockchain)
- LVR: Loss-Versus-Rebalancing
- MDD: Maximum DrawDown
- OLS: Ordinary Least Squares
- PC1: first Principal Component
- PC2: second Principal Component
- PCA: Principal Component Analysis
- SE: Standard Error
- SVB: Silicon Valley Bank
- TAR: Threshold AutoRegression
- TE: Tracking Error
- TMI: Total Market Index (all coins, CRIX methodology)
- TVL: Total Value Locked
- US: United States
- USDC: USD Coin (stablecoin issued by Circle)
- UST: TerraUSD (algorithmic stablecoin of the Terra blockchain)
- XRP: XRP (the native token of the XRP Ledger)

Outline

Motivation

Stylised Facts of Crypto Returns

Crypto Indices: CRIX

Stablecoins

DeFi and Automated Market Makers

Spot ETFs and Crypto Equities

Are Cryptos Becoming Alternative Assets?

Case Study: Risks and Returns of Cryptocurrency

AI for Scientific Discovery

Wrap-up

References

Why Digital Assets in a Course on Market Modelling?

The question of this chapter

- Crypto-assets trade 24/7, have their own indices, stablecoins and exchanges without intermediaries
- Since January 2024 they also trade on US exchanges through spot exchange-traded funds
- Are they a separate asset class, a risk asset like technology shares, or a hedge?

What we do in this chapter

- Measure the market and compare its stylised facts with equities and gold
- Build a crypto index (CRIX) and choose its number of constituents
- Model stablecoins, automated market makers and spot ETFs with real data
- Test whether cryptos are becoming alternative assets

Institutional Background (Recap)



Bust of Satoshi Nakamoto, Graphisoft Park,
Budapest

Photo: Fekist (2021); CC BY-SA 4.0; Wikimedia
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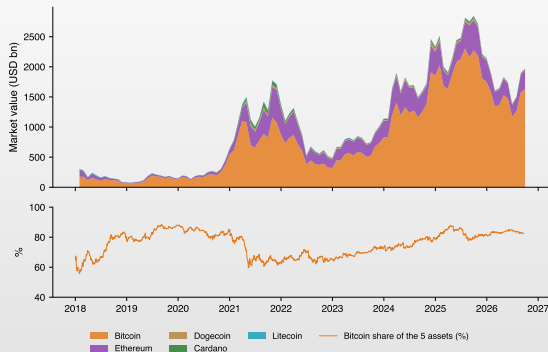
- **Blockchain Nakamoto (2008)**: a shared ledger of blocks, each carrying the hash of the previous one; secured by proof of work (Bitcoin) or proof of stake (Ethereum since 15 September 2022, [ethereum.org: The Merge](https://ethereum.org/en/roadmap/the-merge/))
- Bitcoin issuance halves over time up to 21 million coins [Bitcoin.org FAQ](https://bitcoin.org/en/faq); 20.09 million existed on 18 Sep 2026
- **Coins** are native assets of a blockchain; **tokens** (stablecoins, governance tokens, tokenised funds) are issued by **smart contracts** [Buterin \(2014\)](#); DeFi is built from such contracts [Schär \(2021\)](#)
- Surveys: technology [Lin, Pele & Ren \(2025\)](#); economics [Härdle, Harvey & Reule \(2020\)](#)

Data Used in This Chapter

Series	Content	Period
Crypto prices	Bitcoin, Ethereum, Solana, Dogecoin, Cardano, Litecoin; daily, 7 days a week	2018–2026
Stablecoins	USDT, USDC, DAI prices; supply by coin (DeFiLlama)	2019–2026
Coin supply	Coins in circulation, daily (Coin Metrics Community Data)	2018–2026
DeFi	Total value locked, all chains and by chain (DeFiLlama)	2020–2026
ETFs and shares	spot ETFs on Bitcoin (IBIT) and Ether (ETHA), Coinbase, Strategy; S&P 500, Nasdaq 100, gold	2018–2026
Asset universe	26 assets: cryptos, equity indices, currencies, commodities, bonds	2019–2026

- Daily closes up to 18 September 2026; crypto on its own 7-day calendar; joint analyses on common days (prices joined first, then returns)

Five Coins, One Dominant Asset



- Market value of Bitcoin, Ethereum, Dogecoin, Cardano and Litecoin (month ends) and Bitcoin's share of the five

Reading the Market-Value Chart

- **Market value** $M_{i,t} = P_{i,t} \times Q_{i,t}$ (price in USD times coins in circulation); Bitcoin: 1.63 trillion USD on 18 Sep 2026
 - ▶ Free float matters: XRP and Chainlink report a supply that includes issuer-held tokens, so they are left out of the index universe
- The five coins were worth 1970 billion USD on 18 Sep 2026; peak 3135 billion on 6 Oct 2025
 - ▶ The market value moves with prices, not with supply: supply changes slowly
- Bitcoin's share: 82.5% today, between 55.8% (1 Feb 2018) and 88.5% (5 Sep 2019)
- **Herfindahl–Hirschman index** of concentration: $HHI = \sum_i w_i^2$, with w_i the market-value shares
 - ▶ Today $HHI = 0.71$: the “effective number” of coins $1/HHI$ is only 1.42
- A crypto portfolio weighted by market value is mostly a Bitcoin position

Recap: What Digital Assets Are

- ▣ A blockchain is a shared ledger secured by proof of work or proof of stake
- ▣ Coins are native assets; tokens, stablecoins and DeFi protocols are built by smart contracts
- ▣ Market value = price $\times$ supply in circulation; free float matters
- ▣ The market is highly concentrated: Bitcoin holds about four fifths of our five-coin universe

Notation for This Section

- Daily log return in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$, each series on its own calendar; $m =$ observations per year (about 365 for crypto, 252 for shares); $\sigma_a = s\sqrt{m}$
- Skewness $S = \overline{(r - \bar{r})^3} / s^3$; excess kurtosis $K = \overline{(r - \bar{r})^4} / s^4 - 3$, which estimates a population quantity only if $E|r|^4 < \infty$
- MDD = $\min_t(P_t / \max_{u \leq t} P_u - 1)$; volatility clustering $\hat{\rho}_1(|r|)$ (Chapter 1)
- VaR 1% and ES 2.5% (Chapter 7), historical, daily: $\text{VaR}_{1\%} = -q_{1\%}(r)$, $\text{ES}_{2.5\%} = -\overline{r \mid r \leq q_{2.5\%}(r)}$
- Sample 2018–2026 (Solana from April 2020); comparison assets: S&P 500 and gold

Tail Index: Estimation and Inference

- Hill estimator [Hill \(1975\)](#) from the k largest losses $L_{(1)} \geq \dots \geq L_{(k+1)}$: $\hat{\alpha}_H = \left(\frac{1}{k} \sum_{j=1}^k \ln(L_{(j)}/L_{(k+1)}) \right)^{-1}$, $k = 5\%$ of the losses
 - ▶ Scale-invariant: multiplying returns by a constant leaves $\hat{\alpha}_H$ unchanged; smaller α = heavier tail, $P(L > x) \sim Cx^{-\alpha}$
- Under i.i.d. data with a regularly varying tail, $k \rightarrow \infty$, $k/n \rightarrow 0$, and a second-order condition with negligible bias ($\sqrt{k} A(n/k) \rightarrow 0$): $\sqrt{k}(\hat{\alpha}_H - \alpha) \Rightarrow N(0, \alpha^2)$, so $SE = \hat{\alpha}_H/\sqrt{k}$; the 5% share is a finite-sample tuning choice
 - ▶ With volatility clustering (β -mixing) the variance is larger [Drees \(2000\)](#): use a block bootstrap; choose k from a Hill plot [Drees, de Haan & Resnick \(2000\)](#)
 - ▶ Bitcoin: $\hat{\alpha}_H = 2.73$ ($k = 78$), SE 0.31; with $k = 2.5\%$ or 10% : 3.48 and 2.59, so the choice of k matters as much as the sampling error
- **Consequence:** all six i.i.d. 95% CIs lie below 3.7 (block bootstrap: up to 4.48 for the S&P 500): $\alpha < 4$, hence $E|r|^4 = \infty$, is plausible but not proven
 - ▶ If $\alpha < 4$, the sample kurtosis K estimates nothing: it grows with n and is driven by the largest day; bootstrap CIs for it are invalid (bootstrap failure with infinite variance, [Athreya \(1987\)](#)): report $\hat{\alpha}_H$ with its CI instead

Crypto vs Shares and Gold, 2018–2026

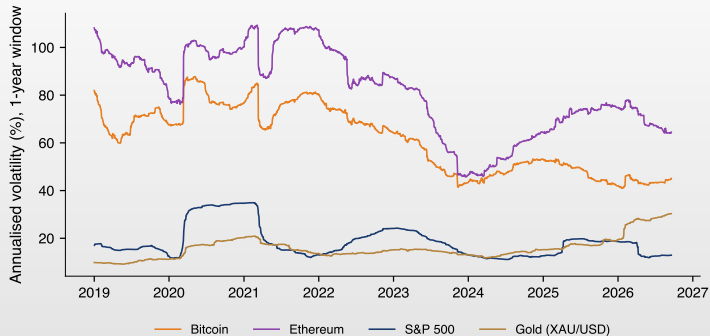
	n	σ_a (%)	S	MDD (%)	$\hat{\rho}_1(r)$	$\hat{\alpha}_H$	95% CI	VaR _{1%}	ES _{2.5%}
Bitcoin	3 182	64	−0.95	−82	0.17	2.73	[2.12, 3.33]	10.2	10.3
Ethereum	3 182	84	−0.82	−94	0.16	2.67	[2.08, 3.26]	13.5	13.7
Solana	2 351	117	−0.22	−96	0.28	2.85	[2.12, 3.58]	14.5	17.1
Dogecoin	3 182	124	4.88	−92	0.37	2.83	[2.22, 3.45]	13.8	16.4
S&P 500	2 189	19	−0.65	−34	0.36	2.87	[2.07, 3.67]	3.4	3.8
Gold	2 269	17	−0.55	−28	0.13	2.75	[1.99, 3.51]	3.0	3.2

- Daily log returns in %; VaR and ES are losses of the daily log return (a log loss ℓ is a loss of $1 - e^{-\ell}$ of the position); $\hat{\alpha}_H$ from the largest 5% of losses, CI $\hat{\alpha}_H \pm 1.96 \hat{\alpha}_H / \sqrt{k}$
- No kurtosis column: if $\alpha < 4$, as the estimates suggest, the sample kurtosis is not a consistent estimate (previous slide)

Reading the Table

- **Volatility:** Bitcoin 64% a year, 3.3 times the S&P 500; Solana and Dogecoin above 100%
 - ▶ Daily VaR 1% of Bitcoin: 10.2% in log-return terms, 3.0 times that of the S&P 500; as a loss of the position: 9.7%
- **Drawdowns:** Bitcoin lost 82% from its peak, Ethereum 94%, Solana 96%; the S&P 500 at most 34%
- **Heavy tails everywhere:** Hill indices between 2.67 and 2.87; the data give no strong evidence of different tail indices
 - ▶ Bitcoin vs S&P 500: $z = -0.28$; Dogecoin vs S&P 500: $z = -0.07$ (i.i.d. SE, independent estimates); crypto has greater volatility, but these tests do not establish equal tail shapes
 - ▶ Overlapping CIs are not a test of equality, and non-rejection is not equivalence: that needs a dependence-aware interval for the difference, robust to the choice of k
- **Dogecoin:** sample kurtosis 106.4 comes from a few social-media-driven jumps; with $\hat{\alpha}_H = 2.83$ a finite population kurtosis is doubtful, so it estimates nothing stable
- Worst Bitcoin day: log return -46.5% (a price fall of -37.2%) on 12 Mar 2020 (COVID-19 crash), four days before the worst S&P 500 day (16 Mar 2020, -12.8%)

Volatility Over Time

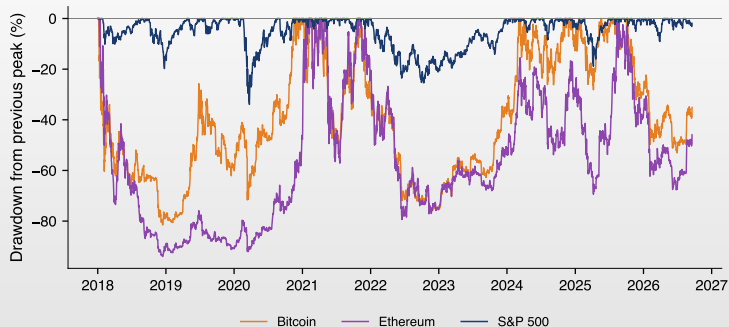


□ Annualised standard deviation of daily log returns over a moving one-year window

Reading the Volatility Chart

- Bitcoin's one-year volatility fell from 88% to 45% today; Ethereum from 109% to 64%
 - ▶ Consistent with a maturing market (deeper liquidity, institutional holders, ETFs), though the chart alone does not identify the cause
- Bitcoin is still 3.5 times as volatile as the S&P 500 (13%)
- Gold's one-year volatility reached 30% in 2026, a record for the sample: the “safe” asset is not always calm
- Ethereum is persistently more volatile than Bitcoin, with a market value about five times smaller

Drawdowns: How Deep and How Long

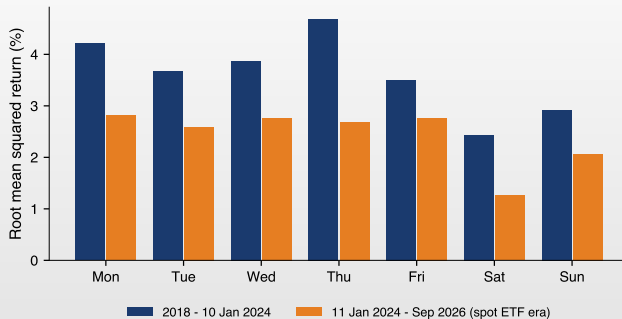


□ Fall from the previous maximum price, in %

Reading the Drawdown Chart

- ▣ Bitcoin was more than 50% below its previous peak on 36% of the days since 2018; Ethereum on 61%
 - ▶ The S&P 500 never fell 50% in this period; its worst drawdown was 34% in March 2020
- ▣ Today: Bitcoin 35%, Ethereum 46% below their peaks
- ▣ Long “crypto winters” (2018–2020, 2022–2023) follow each boom: a reason to study bubbles (Chapter 17)
- ▣ For an investor, drawdown duration matters as much as volatility: capital is tied up for years

A 7-Day Market with a Weekend Lull

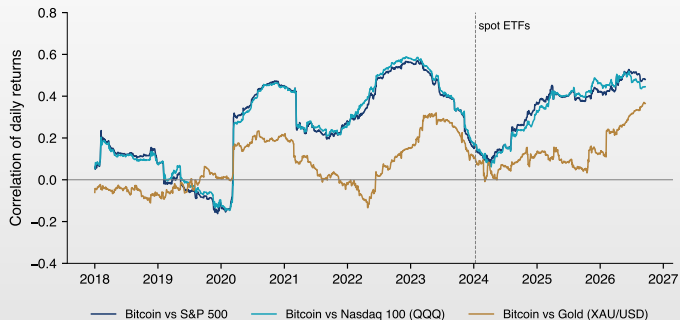


- Root mean squared daily Bitcoin return by day of the week, before and after the spot ETFs started trading

Reading the Day-of-Week Chart

- Weekend variance divided by weekday variance: 0.45 before and 0.39 after 11 January 2024
 - ▶ Bitcoin trades on Saturdays, but the information flow (macro news, US equity trading, ETF flows) does not
- All days became calmer after the ETF launch: weekday root mean square from 4.01% to 2.73%
- Consequence for modelling: annualise crypto volatility with 365 days, but do not treat all days as identical
- Is the weekend ratio different after the ETFs? A block-bootstrap test in Seminar 16 (B1)

Is Bitcoin Moving with Shares?



- Correlation of daily log returns over 250 common trading days (prices joined on common days first)

Reading the Correlation Chart

- Bitcoin vs S&P 500: 0.02 in 2017–2019, 0.39 in 2020–2023, 0.38 since the ETFs (674 days)
 - ▶ The chart suggests a break in early 2020, not in January 2024; the next two slides test it formally
- Nasdaq 100 behaves like the S&P 500 (0.38 since 2024): no special link with technology shares
- Gold: -0.02 , 0.12 , 0.22 : rising, 0.37 in the last window
- Diversification benefit of Bitcoin against shares is smaller than before 2020 [Bouri et al. \(2017\)](#), [Liu & Tsyvinski \(2021\)](#)

Did the Bitcoin–Equity Link Break in 2020? A Formal Test

- Model on 2692 common days since January 2016: $r_t^{\text{BTC}} = a + b r_t^{\text{SPX}} + u_t$, with (a, b) allowed to shift at an unknown date τ
 - ▶ sup-Wald [Andrews \(1993\)](#): $\sup_{\tau \in [0.15T, 0.85T]} W(\tau)$, $W(\tau)$ the Wald statistic for no change, with Newey–West covariances (Bartlett, 8 lags) in each regime
 - ▶ Under H_0 the sup is not χ^2_2 : its limit is $\sup_r \|B(r) - rB(1)\|^2 / [r(1-r)]$ (B a Brownian motion), simulated here: 5% critical value 11.55
- Result: $\sup W = 20.21$ ($p = 0.0008$); least-squares break date 9 Mar 2020 ($W = 18.81$); at the ETF launch $W = 2.31$
 - ▶ 95% CI for the date [Bai \(1997\)](#): 27 Nov 2018 – 15 Jun 2021; it excludes January 2024
 - ▶ Beta to the S&P 500: 0.07 before, 1.22 after; correlation 0.01 and 0.40
- Several breaks at unknown dates: the sequential tests and CIs of [Bai & Perron \(1998\)](#)

Correlation or Volatility? The Forbes–Rigobon Adjustment

- Pearson correlation rises mechanically when the volatility of the driving market rises, with the same underlying link [Forbes & Rigobon \(2002\)](#)
 - ▶ Adjusted correlation: $\rho^* = \rho / \sqrt{1 + \delta(1 - \rho^2)}$, $\delta = \sigma_{\text{SPX,high}}^2 / \sigma_{\text{SPX,low}}^2 - 1$, with 2017–2019 as the calm base period
- 2020–2023: S&P 500 volatility 23.1% vs 12.8% ($\delta = 2.24$); $\hat{\rho} = 0.39$ becomes $\rho^* = 0.23$
 - ▶ Since the ETFs: $\hat{\rho} = 0.38$, $\rho^* = 0.33$ ($\delta = 0.42$)
- Change against 2017–2019 ($\hat{\rho} = 0.02$): $\rho_{2020-23}^* - \hat{\rho}_{2017-19} = 0.21$, 95% block-bootstrap CI [0.09, 0.34]
 - ▶ Under the adjustment's assumptions, 44% of the raw rise is attributed to volatility; the adjusted increase remains positive: stronger co-movement, not an identified mechanism
- Caveat: the adjustment assumes no endogeneity and no omitted common shocks, which a pandemic hardly satisfies

Recap: Stylised Facts

- ▣ Crypto is 3–7 times as volatile as shares, with drawdowns of 80–96%
- ▣ Heavy tails with Hill indices near 2.7 for crypto, shares and gold alike (overlapping CIs), and volatility clustering
- ▣ Volatility has fallen over time; weekends are calmer than weekdays
- ▣ A tested break in March 2020 (not January 2024): since then Bitcoin co-moves more with shares, beyond the volatility effect; the break dates the change, it does not explain it

Why a Crypto Index?



Main building of Humboldt-Universität zu Berlin, where
CRIX was developed

Photo: [Diego Delso \(2016\)](#); CC BY-SA 4.0; Wikimedia

[Commons](#)

- A benchmark for fund managers and a basis for investable products
- **CRIX** [Trimborn & Härdle \(2018\)](#): a market-value weighted index of the crypto market
 - ▶ The number of constituents is **not fixed**: it is chosen by the Akaike information criterion (AIC) so that the index tracks the whole market
 - ▶ Reason: coins appear and vanish daily; a fixed number (like 30 or 500 for shares) would be arbitrary
- Published at thecrix.de; a review of crypto indices: [Găman, Pele, Härdle & Będowska-Sójka \(2026\)](#)

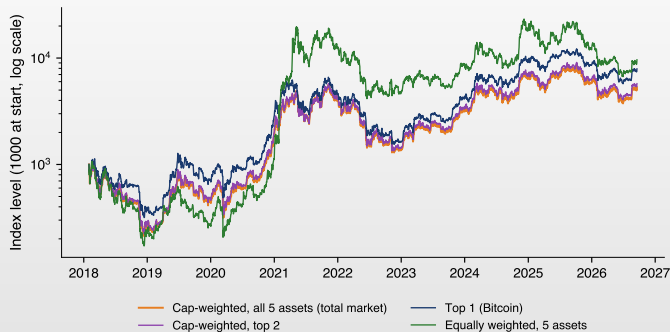
A Market-Value Weighted (Laspeyres) Index

- Level on day t in reallocation period τ : $I_t = \frac{\sum_{i=1}^k P_{i,t} Q_{i,\tau}}{D_\tau}$
 - ▶ $Q_{i,\tau}$: coins in circulation, fixed during the period; D_τ : the **divisor**
- Equivalent form: the index return is the weighted average of the constituents' returns
 - ▶ $\frac{I_t}{I_{t_0}} = \sum_i w_i \frac{P_{i,t}}{P_{i,t_0}}$, $w_i = \frac{P_{i,t_0} Q_{i,\tau}}{\sum_j P_{j,t_0} Q_{j,\tau}}$ (market-value shares at the start of the period)
- At a **reallocation** (new Q , new members) the divisor changes so that the level does not jump
 - ▶ $D_{\tau+1} = \frac{\sum_i P_{i,t} Q_{i,\tau+1}}{I_t}$: old and new baskets give the same I_t on the switch day
- In this chapter: monthly reallocation, weights from the market values at the previous month end
 - ▶ Worked examples with a divisor and a constituent change: Seminar 16, A1–A2

How Many Constituents? The Published CRIX Rule

- Total market index TMI (all coins) and candidate indices $\text{CRIX}(k, \beta)$: the k_1 largest coins at weight 1, the next $s = k - k_1$ with free weights β (Trimborn & Härdle (2018), Section 3, eqs. (11)–(12) in Trimborn & Härdle (2018, author version))
 - ▶ $\hat{\varepsilon}_t(k, \beta) = \Delta \ln \text{TMI}_t - \Delta \ln \text{CRIX}_t(k, \beta)$; $\hat{\beta}$ minimises $\|\hat{\varepsilon}(k, \beta)\|^2$
- $\text{AIC}(k) = -2 \log L\{\hat{\varepsilon}(k, \hat{\beta})\} + 2s$ (eq. (20)); L uses an Epanechnikov kernel density of $\hat{\varepsilon}(k_1, 1)$ with a Sheather–Jones plug-in bandwidth
 - ▶ Under Normal errors it reduces to $T \ln \hat{\sigma}_k^2 + 2s$ (Appendix, eq. (35))
- ECRIX: $k_1 = 1$, steps of one coin, stop at the first rise of the AIC (Section 5, rule (26)); EFCRIX: global minimum (rule (27))
 - ▶ Members reallocated monthly by market value; k reviewed quarterly on the last three months of data (Section 4)
- Data limitation: circulating supply is public for five of our coins only (Bitcoin, Ethereum, Dogecoin, Cardano, Litecoin), so $k \leq 4$ and TMI has five members, not hundreds

Crypto Indices Since 2018

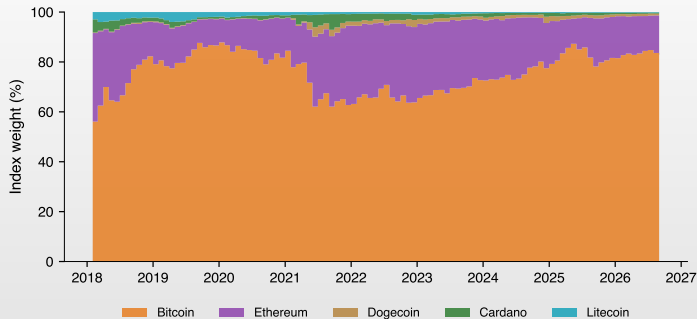


- Monthly reallocation; weights from market values at the previous month end; log scale

Reading the Index Chart

- Total-market index: 21.4% a year over 8.6 years, volatility 65.7%
 - ▶ The compound annual growth rate is $(I_T/I_0)^{1/\text{years}} - 1$
- Bitcoin alone did better than the market-value index: the other coins lost weight and value
- Equal weights: 30.0% a year, but volatility 82.5%
 - ▶ Driven by one year: 2021 (+998% vs +113%), when Dogecoin and Cardano soared; a few extreme months, not a stable premium
- An index is only as good as its universe: survivors today are not the leaders of 2018 [Będowska-Sójka, Wójcik & Pele \(2026\)](#)

Index Weights Over Time

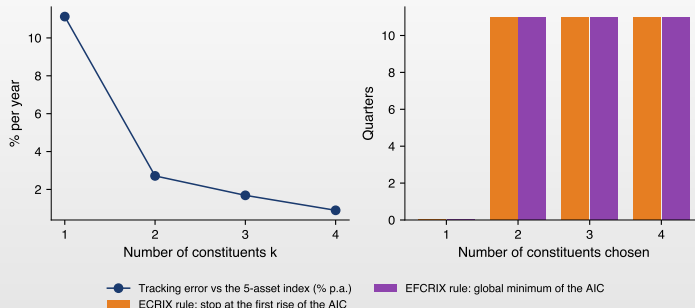


□ Market-value weights at each monthly reallocation, in %

Reading the Weights

- ▣ Bitcoin's weight: 75.7% on average, never below 56.1%; today 82.9%
- ▣ Ethereum 15.8%; Dogecoin, Cardano and Litecoin together below 2%
- ▣ The 2021 bull market briefly lifted the smaller coins, then Bitcoin regained its share
 - ▶ Altcoins (coins other than Bitcoin) gain weight late in booms and lose it in busts
- ▣ Concentration makes crypto indices behave like Bitcoin with a small add-on

Tracking Error and the CRIX Rule



- Left: tracking error of the top- k index (weights 1) against the five-coin index; right: number of quarters, 33 in total, in which each rule chooses k constituents

Reading the CRIX-Rule Chart

- Bitcoin alone tracks the five-coin market with an error of 11.1% a year; the top two with 2.7%, the top three 1.7%, the top four 0.9%
 - ▶ When coins are ranked by market value, the error falls fast: most of the market is in the first two coins
- The published rule chooses 2 coins in 11, 3 in 11 and 4 in 11 of 33 quarters; 2 in the last one
 - ▶ ECRIX and EFCRIX agree in every quarter: the AIC never rises before its global minimum
 - ▶ Free weights β let a small coin proxy for the rest, and three-month windows make the choice adapt to the market
- CRIX applies the same rule to hundreds of coins, where the smallest ones barely move the total [Trimborn & Härdle \(2018\)](#)
- Project idea: extend the universe with coins whose circulating supply is public and redo the selection

Recap: Crypto Indices

- ▣ A Laspeyres index with a divisor: fixed quantities within a period, continuity at reallocation
- ▣ CRIX chooses its number of constituents by an AIC on tracking residuals, reviewed quarterly; on our five coins it picks 2–4
- ▣ Market-value weighting makes crypto indices mostly Bitcoin; equal weights add risk, not a stable premium
- ▣ Free float and survivorship are the two practical traps

Three Ways to Keep a Token at 1 USD

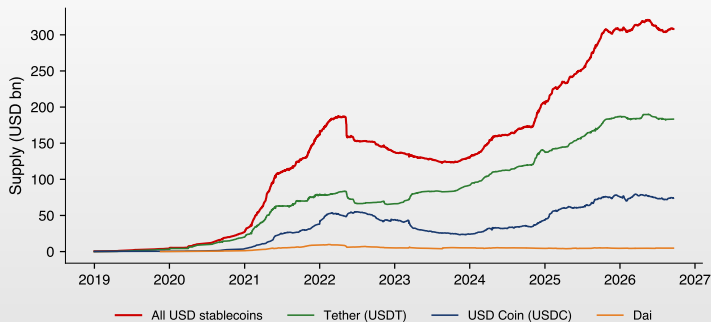
Design	Backing	Examples (USD bn, 18 Sep 2026)
Fiat-backed	Bank deposits and short US Treasury bills held by the issuer	USDT 183.33, USDC 73.92, PYUSD 2.81
Crypto-backed	Over-collateralised crypto deposits or hedged positions, managed by smart contracts	USDS 6.49, USDe 4.74, DAI 4.77
Algorithmic	No external collateral: supply expands and contracts against a second token	TerraUSD (collapsed in May 2022)

- All USD stablecoins on 18 Sep 2026: 308 billion USD ([DefiLlama](#)); by design, the 45 coins above 100 million USD
 - Fiat-backed 283 billion, crypto-backed 22.4 billion, algorithmic 0.15 billion; all smaller coins together 2.6 billion
- Regulation: reserves, redemption and disclosure rules in [MiCA](#), [Regulation \(EU\) 2023/1114](#) and [GENIUS Act](#), [Public Law 119-27](#)

Keeping the Peg: Arbitrage Bands

- Authorised traders can **mint** (pay 1 USD, receive 1 coin) and **redeem** (return 1 coin, receive 1 USD) with the issuer
 - ▶ With a redemption cost c_r and a minting cost c_m (fees, settlement time), arbitrage keeps the market price P in the band $1 - c_r \leq P \leq 1 + c_m$
- Below the band: buy on the market, redeem at 1 USD; above: mint at 1 USD, sell on the market
- The band breaks when redemption stops (bank holiday, frozen reserves) or when the backing is doubted
 - ▶ Evidence on arbitrage and issuance: [Lyons & Viswanath-Natraj \(2023\)](#); on the link between Tether issuance and Bitcoin prices: [Griffin & Shams \(2020\)](#)
- Band dynamics step by step: Seminar 16, A7–A8

The Rise of Stablecoins



Supply in circulation, billion USD (DefiLlama)

MFM_ch16_stablecoins

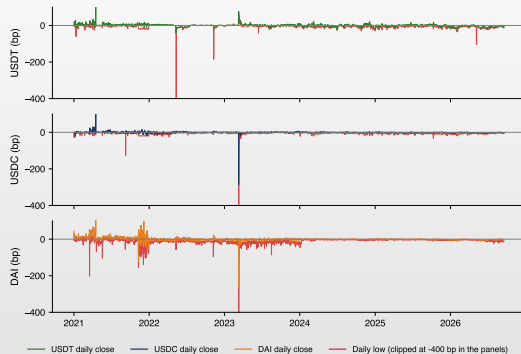
Reading the Supply Chart

- ▣ From 4.2 billion USD in January 2020 to 308 billion on 18 Sep 2026: about 74 times
 - ▶ Stablecoins are the cash leg of crypto trading and DeFi, and a channel for dollar payments
- ▣ Tether: 183 billion (60% of the total); USD Coin: 74 billion (24%); Dai: 4.8 billion
- ▣ The 2022–2023 dip: the total fell from 187 billion (2 Apr 2022) to 123 billion (19 Aug 2023); TerraUSD vanished and USD Coin fell from 55 to 23 billion
- ▣ Reserves of payment stablecoins sit in cash-like assets such as short-term Treasury bills ([GENIUS Act, Public Law 119-27](#)): stablecoins now matter for the money market

Arbitrage Bands Imply a Threshold AR

- Deviation from the peg in basis points (1 bp = 0.01%): $d_t = 10^4(P_t - 1)$, daily close
 - ▶ Inside the band $|d| \leq c$ arbitrage does not pay: weak pull; outside it does: fast reversion
- Threshold autoregression (TAR), the equilibrium form of [Balke & Fomby \(1997\)](#): $d_t = \phi_{\text{in}} d_{t-1} + \varepsilon_t$ if $|d_{t-1}| \leq c$, $d_t = \phi_{\text{out}} d_{t-1} + \varepsilon_t$ otherwise
 - ▶ $\hat{c}$ by least squares over a grid of $|d_{t-1}|$ values, keeping at least 15% of the days in each regime
 - ▶ $\hat{c}$ (in bp) is an empirical regime boundary: it matches the costs only under a symmetric band $c_r = c_m$ (USD per token), as $10^4 c_r$ bp, and other frictions also move it
- Test of linearity $H_0: \phi_{\text{in}} = \phi_{\text{out}}$: c is not identified under H_0 , so $\sup_c W(c)$ is not χ_1^2 (the Davies problem)
 - ▶ p -value from the fixed-regressor bootstrap of [Hansen \(1996\)](#) with heteroskedasticity: $d_t^* = \hat{\varepsilon}_t \eta_t$, $\eta_t \sim N(0, 1)$, regressors held fixed; $W(c)$ with White standard errors
- Sample 2021–2026: earlier quotes are thin and noisy

Deviations from 1 USD



- Daily close and daily low minus 1 USD, in basis points, 2021–2026; lows below -400 bp are cut at the panel edge

Reading the Peg Chart

	s.d. (bp)	$ d_t > 50$ (%)	lowest close (bp)	lowest low (bp)
Tether	7.3	0.14	-41	-515
USD Coin	7.5	0.14	-285	-1226
Dai	10.8	0.57	-261	-1030

- In normal times the peg holds within a few basis points
- Stress is rare but deep: USD Coin's daily low reached -1226 bp on 11 Mar 2023; Tether's -515 bp on 12 May 2022
- Holding a stablecoin is like holding a claim with a short put on the reserves: nothing happens most days, a large loss is possible on a few

Threshold AR Estimates for the Three Stablecoins

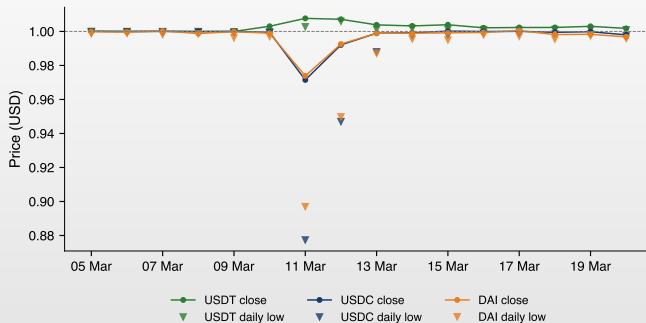
	$\hat{c}$ (bp)	$\hat{\phi}_{in}$ (SE)	$\hat{\phi}_{out}$ (SE)	sup W	p bootstrap	p (χ^2_1)	$\hat{\phi}$ linear
Tether	8.6	0.86 (0.02)	0.69 (0.11)	7.4	0.084	0.006	0.73
USD Coin	2.9	0.59 (0.04)	0.28 (0.03)	32.5	0.001	< 0.001	0.29
Dai	6.0	0.72 (0.08)	0.32 (0.07)	13.8	0.005	< 0.001	0.33

- USD Coin and Dai: linearity rejected; reversion is much faster outside the band (half-life 0.5 vs 1.3 days for USD Coin)
 - ▶ Without 9–16 March 2023, USD Coin gives $p = 0.29$ (bootstrap) but 0.019 (χ^2_1): one stress week carries the evidence, and χ^2 critical values over-reject
- Tether: the widest band ($\hat{c} = 8.6$ bp) and slow reversion inside it, but linearity is not rejected at 5%
 - ▶ Consistent with fewer direct redeemers: who arbitrages the peg matters [Ma, Zeng & Zhang \(2025\)](#)
- For Tether and USD Coin $\hat{c}$ lies at the upper end of the trimmed grid: the band edge is far in the tail of $|d_t|$

March 2023: a Bank Failure Hits a Stablecoin

- Friday 10 March 2023: Silicon Valley Bank (SVB) is closed and the Federal Deposit Insurance Corporation (FDIC) is appointed receiver [FDIC \(2023\)](#)
 - ▶ USD Coin relied on reserves held at SVB, so holders feared a loss on the backing [Diop, Chevallier & Sanhaji \(2024\)](#)
- Weekend: banks closed, redemptions could not settle, so the arbitrage band disappeared
- Sunday 12 March: the Treasury, the Federal Reserve and the FDIC announce that all SVB depositors are protected, with access from Monday [Treasury, Federal Reserve & FDIC \(2023\)](#)
- The peg returned within two days: the event tests the backing, not the design

USD Coin and Dai Below Par, Tether Above



□ Daily close (lines) and daily low (triangles), 5–20 March 2023

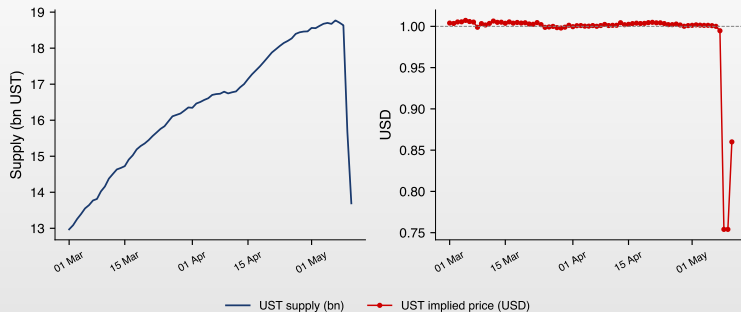
Reading the March 2023 Chart

- ▣ USD Coin: daily low 0.8774 USD on Saturday 11 March, close 0.9715
 - ▶ Dai fell with it (low 0.8970): a crypto-backed coin hit through its collateral [Diop, Chevallier & Sanhaji \(2024\)](#)
- ▣ Tether rose to 1.0077 at the close (high 1.0296): a flight to the other large stablecoin
- ▣ Contagion from traditional banking to crypto, the reverse of the usual worry
- ▣ For risk models: peg deviations have extremely heavy tails; a VaR from normal days is meaningless here

TerraUSD: an Algorithmic Stablecoin Collapses

- TerraUSD (UST) kept its peg by exchange with the Terra token LUNA, not with reserves
 - ▶ Demand came from a lending protocol (Anchor) that promised high yields to UST depositors [Liu, Makarov & Schoar \(2023\)](#)
- Terra, the third largest crypto ecosystem after Bitcoin and Ethereum, collapsed in three days in May 2022 and wiped out 50 billion USD in valuation [Liu, Makarov & Schoar \(2023\)](#)
- After a few large holders moved on 7 May, others followed: a bank run, visible on the blockchain in real time
- Wealthier, more sophisticated investors ran first and lost less [Liu, Makarov & Schoar \(2023\)](#)

The Run on TerraUSD



□ Supply and implied price (market value / supply), March–May 2022 ([DefiLlama](#)); the series ends on 11 May 2022

Reading the TerraUSD Chart

- ▣ Supply grew steadily to 18.8 billion on 7 May 2022; four days later it was 13.7 billion: 27% of the coins were redeemed
- ▣ Implied price fell to 0.754 USD on 10 May 2022; redemption into LUNA created more LUNA and pushed its price down: a death spiral
- ▣ Without external collateral, the peg rests on confidence alone: a run is self-fulfilling
- ▣ Lesson for regulators: payment stablecoins must hold liquid reserves ([GENIUS Act](#), [Public Law 119-27](#), [MiCA](#), [Regulation \(EU\) 2023/1114](#))

Tokenisation of Real-World Assets

- ▣ **Tokenised yield-bearing instruments:** claims on Treasury bills and cash issued as tokens on a blockchain; the legal form differs
 - ▶ Fund shares: BlackRock BUIDL 2.38, Circle USYC 2.55; a tokenised secured note issued by a special-purpose vehicle: Ondo USDY 2.22 ([Ondo](#)) (billion USD, 18 Sep 2026, [DefiLlama](#))
- ▣ Unlike a payment stablecoin, they pay the Treasury yield to the holder
- ▣ Benefits: settlement 24/7, use as collateral in DeFi; risks: smart-contract bugs, legal ownership, liquidity in stress [Aramonte, Huang & Schrimpf \(2021\)](#)
- ▣ Token economics and adoption: [Cong, Li & Wang \(2021\)](#)

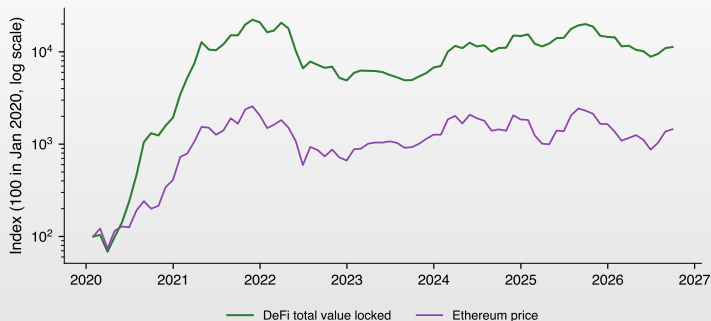
Recap: Stablecoins

- ▣ Three designs: fiat-backed, crypto-backed, algorithmic; fiat-backed dominate
- ▣ The peg is an arbitrage band that holds only while redemption works
- ▣ A threshold AR fits the band: slow drift inside, fast reversion outside; test linearity with a bootstrap, not with χ^2
- ▣ Two runs: USD Coin in March 2023 (a reserve shock, recovered), TerraUSD in May 2022 (no reserves, collapsed)

Decentralised Finance in Five Layers

- Schär (2021) describes DeFi as a stack of layers
 - ▶ **Settlement**: the blockchain and its native coin (Ether)
 - ▶ **Assets**: tokens issued on it (stablecoins, governance tokens)
 - ▶ **Protocols**: exchanges, lending, derivatives as smart contracts
 - ▶ **Applications and aggregators**: user interfaces that route trades across protocols
- Promise: open access, transparency, composability (protocols plug into each other)
- Risks: high leverage, liquidity mismatches, built-in interconnectedness, no shock absorber Aramonte, Huang & Schrimpf (2021)
 - ▶ “Decentralisation illusion”: governance tends to concentrate power Aramonte, Huang & Schrimpf (2021)

Total Value Locked in DeFi

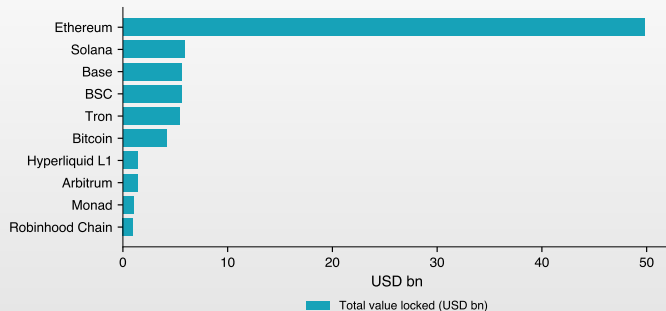


□ DeFi total value locked, all chains ([DeFiLlama](#)), and the Ethereum price, month ends, both 100 in January 2020, log scale

Reading the TVL Chart

- TVL: 0.6 billion USD in January 2020, 177 billion at the peak (9 Nov 2021), 88 billion on 18 Sep 2026 (50% below the peak)
 - ▶ The “DeFi summer” of 2020 multiplied TVL far faster than the Ether price
- Monthly log changes: $\Delta \ln \text{TVL}_t = a + b \Delta \ln P_t^{\text{ETH}} + u_t$ with Newey–West standard errors
 - ▶ $\hat{b} = 0.79$ (SE 0.07), $R^2 = 0.62$, $n = 80$ months
- TVL co-moves with Ether prices; this regression does not separate valuation changes from net deposits, which can move together with the price
- Cleaner activity measures: token quantities valued at fixed prices, net flows, or the fees paid by users; dividing by the Ether price still leaves the valuation of other tokens and stablecoins

Where the Value Is Locked



□ Total value locked by blockchain, the ten largest, 18 September 2026 ([DefiLlama](#))

Reading the Chain Chart

- ▣ Ethereum holds 49.8 billion USD, 57% of the 88 billion across all chains on the same date (the end point of the TVL chart)
- ▣ Next: Solana 5.9, Base 5.6, BSC (BNB Smart Chain) 5.6, Tron 5.4, Bitcoin 4.1 billion
- ▣ Base and Arbitrum are “layer-2” networks that settle on Ethereum: cheaper transactions, same security anchor
- ▣ Concentration again: one settlement layer carries most of DeFi

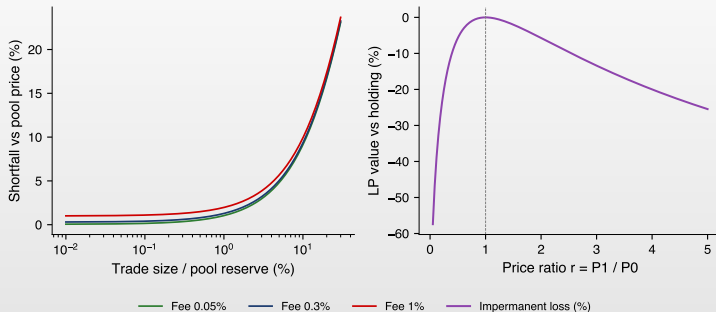
The Constant-Product Market Maker

- A pool holds reserves x of token X (e.g. Ether) and y of token Y (e.g. USD Coin); it accepts any trade that keeps $x \cdot y = k$ [Adams, Zinsmeister & Robinson \(2020\)](#)
 - ▶ Pool price (marginal): $p = y/x$ units of Y per X
- A trader sells Δx ; the pool keeps a fee f , so $\gamma \Delta x$ enters the invariant, $\gamma = 1 - f$
 - ▶ $(x + \gamma \Delta x)(y - \Delta y) = xy \Rightarrow \Delta y = \frac{y \gamma \Delta x}{x + \gamma \Delta x}$
- Execution price $\Delta y / \Delta x = p \frac{\gamma}{1 + \gamma \delta}$, with $\delta = \Delta x / x$ the trade size relative to the reserve
 - ▶ **Shortfall** against the pool price: $1 - \gamma / (1 + \gamma \delta)$: fee plus price impact
- Arbitrageurs keep p in line with other exchanges: the pool price follows the market [Angeris et al. \(2019\)](#)

Worked Example: a Swap in an Ether Pool

- Pool: $x = 2\,000$ Ether, $y = 5\,000\,000$ USD Coin, so $p = 2\,500$ and $k = 10\,000 \times 10^6$; fee $f = 0.3\%$
 - ▶ Sell $\Delta x = 20$ Ether ($\delta = 1\%$); after the fee, $\gamma \Delta x = 19.94$
- $\Delta y = \frac{5\,000\,000 \times 19.94}{2\,000 + 19.94} = 49\,358$ USD Coin, an execution price of 2467.9 per Ether
- Shortfall: $1 - 2467.9/2\,500 = 1.28\%$ (0.3% fee plus about 1% price impact)
- New pool price: $(5\,000\,000 - \Delta y)/2\,020 = 2450.8$, i.e. -1.97%
 - ▶ An arbitrageur now buys Ether in the pool and sells it elsewhere until p is back at the market price

Slippage and Impermanent Loss



- Left: shortfall against the pool price by trade size and fee; right: value of a liquidity position relative to holding the two tokens

Reading the Slippage Panel

- With a 0.3% fee: shortfall 0.40% for a trade of 0.1% of the reserve, 1.28% for 1%, 9.34% for 10%
 - ▶ Small trades pay the fee; large trades pay the price impact, roughly δ for small δ
- Lower fee tiers (0.05%) suit stable pairs; at $\delta = 1\%$ the shortfall is 1.04%
- Deep pools (large x) mean small δ : liquidity is measured by the reserve, not by an order book (Chapter 10)
- Concentrated liquidity (Uniswap v3) lets providers place reserves in a price range, deepening the pool near the current price (v3 white paper)

Impermanent Loss, Step by Step

- A **liquidity provider** (LP) deposits x_0 and $y_0 = p_0 x_0$ (half and half in value) and receives a share of the pool
 - ▶ Without fees, arbitrage moves the pool to the market price p_1 : from $xy = k$ and $y/x = p_1$, $x_1 = \sqrt{k/p_1}$, $y_1 = \sqrt{k p_1}$
- LP value: $V_{LP} = p_1 x_1 + y_1 = 2\sqrt{k p_1}$; holding the tokens: $V_H = p_1 x_0 + y_0$
- With $r = p_1/p_0$ and $k = p_0 x_0^2$: **impermanent loss** $IL(r) = \frac{V_{LP}}{V_H} - 1 = \frac{2\sqrt{r}}{1+r} - 1 \leq 0$
 - ▶ $r = 2$ or $r = 0.5$: -5.72%; $r = 4$ or 0.25 : -20.00%; symmetric in $\ln r$
- “Impermanent”: it vanishes if the price returns to p_0 ; it becomes permanent when the LP withdraws

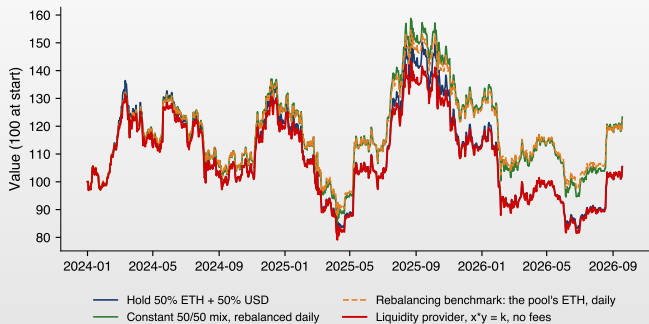
Loss Versus Rebalancing

- A fairer benchmark than holding: a **rebalancing** portfolio that holds the pool's quantity of Ether at every instant and trades at market prices [Milionis et al. \(2022\)](#)
 - ▶ The pool rebalances too, but at stale prices set by arbitrageurs: it always trades at a loss to them
- For $x \cdot y = k$ and a price with volatility σ , the loss versus rebalancing (LVR) grows at the rate
 - ▶ $\frac{\text{LVR}}{V} = \frac{\sigma^2}{8}$ per unit of time [Milionis et al. \(2022\)](#)
- Fees must exceed $\sigma^2/8$ a year for a liquidity provider to beat this benchmark (not to earn a positive return): a volatility-selling business
- Next: the derivation, then an Ether/USD pool since January 2024, without fees

Deriving $LVR = \sigma^2/8$

- Pool value at market price P after arbitrage: $V(P) = Px + y = 2\sqrt{kP}$, concave: $V'(P) = \sqrt{k/P} = x$,
 $V''(P) = -\frac{1}{2}\sqrt{k}P^{-3/2}$
 - ▶ Rebalancing portfolio: holds the same $x_t = V'(P_t)$ Ether at every instant, so $dR_t = V'(P_t) dP_t$
- Price $dP_t = \mu P_t dt + \sigma P_t dW_t$; Itô: $dV_t = V' dP_t + \frac{1}{2} V'' \sigma^2 P_t^2 dt$
 - ▶ $d(R_t - V_t) = -\frac{1}{2} V''(P_t) \sigma^2 P_t^2 dt = \frac{1}{4} \sqrt{kP_t} \sigma^2 dt = \frac{\sigma^2}{8} V_t dt$
- LVR is the pool's negative gamma: a short-volatility position that loses $\sigma^2/8$ of its value per unit of time, whatever the drift [Milionis et al. \(2022\)](#)
- Assumptions: continuous geometric Brownian motion, frictionless arbitrage, no fees, continuous blocks
 - ▶ With fees and discrete blocks, arbitrageurs trade only when the price leaves the fee band, which lowers the loss [Milionis, Moallemi & Roughgarden \(2025\)](#)
 - ▶ Weighted pool $x^w y^{1-w} = k$: rate $\sigma^2 w(1-w)/2$ (Seminar 16, A6)

A Liquidity Provider in an Ether Pool



- Daily Ether prices from 1 January 2024; 100 invested half in Ether, half in USD; pool with no fees; dashed: the LVR benchmark (the pool's Ether, rebalanced daily)

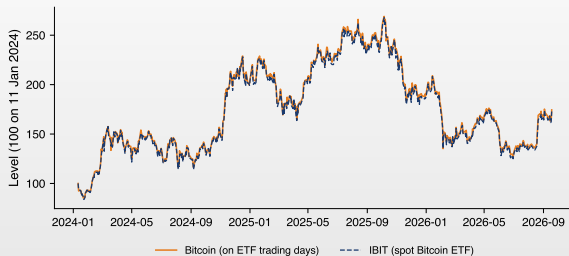
Reading the Liquidity-Provider Chart

- ▣ Ether: 2352 USD at the start, 2611 at the end ($r = 1.11$): the impermanent loss is only -0.14%
 - ▶ Final values: holding 105.5, liquidity provider 105.4, LVR benchmark 122.4 (cumulative LVR 17.0), constant 50/50 mix 123.3
- ▣ Realised LVR rate $\sum_t (\Delta R_t - \Delta V_t) / V_{t-1}$ per year: 5.83%; theory: $\sigma^2/8 = 0.463/8 = 5.79\%$ with $\sigma = 68.1\%$
 - ▶ 95% block-bootstrap CI for the realised rate: [4.88, 6.98]% a year; 2.7 years pin it down only to about ± 1 point
 - ▶ The match is not a test: each day's loss is $R_t/2 - (\sqrt{1 + R_t} - 1) \approx R_t^2/8$, so the realised LVR is the realised variance over 8 (bootstrap correlation of the two: 0.9981)
 - ▶ The constant 50/50 mix is a different benchmark with the same instantaneous rate: its log growth exceeds the pool's by $\sigma^2/8$, here 5.79% a year
- ▣ Impermanent loss is small when the price returns; LVR accumulates with every price move
- ▣ Fee income needed to match the benchmark: about 5.79% of the position a year in this period

Recap: DeFi and AMMs

- DeFi is a stack of smart contracts on a settlement blockchain; its risks are leverage, interconnection and governance
- TVL follows the Ether price: read it in units of Ether, not only in USD
- Constant product $xy = k$: execution price $p\gamma/(1 + \gamma\delta)$; impermanent loss $2\sqrt{r}/(1 + r) - 1$
- Providing liquidity is selling volatility: fees must beat $\sigma^2/8$ a year

IBIT Follows Bitcoin

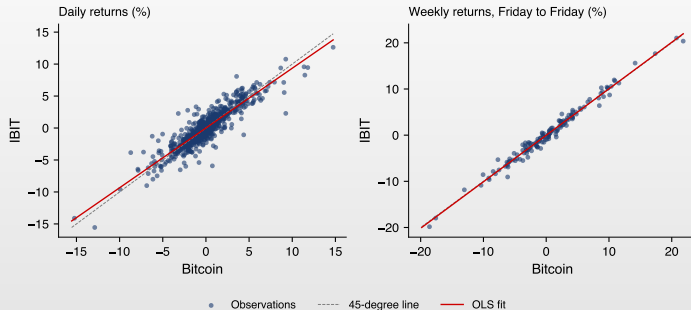


- Bitcoin and IBIT on ETF trading days, both 100 on 11 Jan 2024
- A **spot ETF** holds the coins with a custodian and trades in US hours; BlackRock's IBIT charges a sponsor fee of 0.25% a year [BlackRock \(2026\)](#)
- **Authorised participants** create or redeem shares against coins, which keeps the price within a cost band around the net asset value, as for a stablecoin

Measuring How Well an ETF Tracks

- Regression of ETF on coin returns, on the ETF's trading days: $r_t^{\text{ETF}} = a + \beta r_t^{\text{coin}} + u_t$
 - ▶ A perfect tracker: $\beta = 1$, $R^2 = 1$
- **Tracking error:** $\text{TE} = \text{sd}(r_t^{\text{ETF}} - r_t^{\text{coin}})\sqrt{m}$, with $m = 252$ (daily) or 52 (weekly)
- **Fee drift:** slope of $\ln(P_t^{\text{ETF}}/P_t^{\text{coin}})$ on time in years
 - ▶ The fund sells a little Bitcoin every day to pay its fee: the coins per share shrink at the fee rate
- **Timing:** the coin's daily close and the ETF's close are recorded at different times of the day

Daily vs Weekly Returns



□ IBIT against Bitcoin: daily returns on common days (left) and Friday-to-Friday returns (right)

Reading the Tracking Results

	$\hat{\beta}$ daily	R^2 daily	$\hat{\beta}$ weekly	R^2 weekly	TE daily	TE weekly
IBIT / Bitcoin	0.93	0.82	1.01	0.98	20.8	6.2
ETHA / Ether	0.94	0.83	0.99	0.98	29.9	9.0

- Daily data understate tracking ($R^2 = 0.82$, TE 20.8%); weekly data show a near-perfect tracker ($R^2 = 0.98$, TE 6.2%)
 - ▶ The daily gap is a timing mismatch between two closing prices, which averages out over a week
- Fee drift: -0.24% a year for IBIT and -0.24% for ETHA, close to the 0.25% fee
- IBIT trades 2.2 billion USD a day on average (maximum 10.3 billion on 5 Feb 2026)

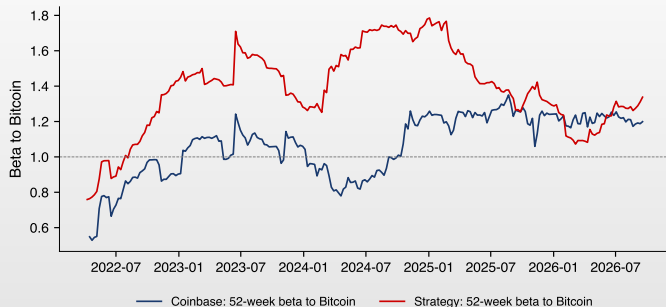
Nonsynchronous Closes: the Dimson Beta

- The coin's daily close and the ETF's New York close are hours apart: part of day t 's coin move reaches the ETF on day $t \pm 1$
 - ▶ Daily OLS beta is then biased towards zero; Newey–West fixes the standard error, not the bias
- Dimson (1979): $r_t^{\text{ETF}} = a + \sum_{j=-1}^1 b_j r_{t+j}^{\text{coin}} + u_t$ on common days; sum beta $\hat{\beta}_D = \hat{b}_{-1} + \hat{b}_0 + \hat{b}_1$
 - ▶ IBIT: $\hat{b}_{-1} = 0.05$, $\hat{b}_0 = 0.94$, $\hat{b}_1 = 0.01$; $\hat{\beta}_D = 1.00$ (SE 0.023) vs OLS 0.94
 - ▶ ETHA: $\hat{\beta}_D = 0.99$ (SE 0.022) vs OLS 0.94; both sum betas are within one SE of 1
- The daily Dimson beta reaches what the weekly regression shows (1.01) with five times more observations
- Where price discovery happens (spot or ETF) is a separate question: information shares [Hasbrouck \(1995\)](#)

Crypto Equities: Coinbase and Strategy

- **Coinbase:** a US crypto exchange; revenue depends on trading volume, which rises with prices
 - ▶ **Strategy** (formerly MicroStrategy): a software company that holds Bitcoin as its main treasury asset
- Weekly two-factor regression with Newey–West standard errors, April 2021 – September 2026
 - ▶ $r_t^{\text{stock}} = a + \beta_{\text{BTC}} r_t^{\text{BTC}} + \beta_{\text{SPX}} r_t^{\text{S\&P}} + u_t$
- Weekly returns avoid the timing mismatch between the 7-day crypto market and the 5-day stock market
- A stock with $\beta_{\text{BTC}} > 1$ is a leveraged Bitcoin position with extra equity risk

Beta to Bitcoin Over Time



□ Rolling 52-week beta of weekly returns on Bitcoin weekly returns

Reading the Crypto-Equity Results

	$\hat{\beta}_{\text{BTC}}$	SE	$\hat{\beta}_{\text{SPX}}$	SE	R^2	σ_a (%)
Coinbase	0.79	0.09	2.05	0.22	0.52	84
Strategy	1.18	0.11	1.02	0.24	0.63	89

- Strategy moves 1.18 times Bitcoin plus 1.02 times the S&P 500; its rolling beta ranged from 0.76 to 1.79
- Coinbase is more an equity than a Bitcoin proxy: $\hat{\beta}_{\text{SPX}} = 2.05$, a high-beta US stock whose business depends on crypto activity
- Both are more volatile than Bitcoin itself (55% a year on the same weeks)

Recap: ETFs and Crypto Equities

- ▣ Spot ETFs track the coin almost perfectly once timing is handled (weekly R^2 about 0.98)
- ▣ The cost is a fee drift of about 0.25% a year, visible directly in the price ratio
- ▣ Daily correlations between assets traded at different hours are biased downwards: use weekly returns
- ▣ Crypto equities mix Bitcoin exposure with equity-market risk and company-specific leverage

Crypto Factors: What Prices the Cross-Section?

- [Liu & Tsyvinski \(2021\)](#): crypto returns have little exposure to stock, currency or commodity factors; they load on crypto momentum and investor attention
 - ▶ [Liu, Tsyvinski & Wu \(2022\)](#): three factors from weekly coin returns: CMKT (value-weighted market excess return), CSIZE (small minus big coins), CMOM (past three-week winners minus losers)
- Value from on-chain adoption and segmentation from traditional markets [Cong, Karolyi, Tang & Zhao \(2026\)](#); mispricing versus risk compensation [Babiak & Bianchi \(2025\)](#)
- Fundamentals: network value follows Metcalfe's law in the long run, not in the short run [Pele & Mazurencu-Marinescu-Pele \(2019a\)](#); adoption and token value reinforce each other [Cong, Li & Wang \(2021\)](#)
- Two different questions, tested separately on the next slides
 - ▶ Model adequacy: are all test-asset alphas zero for the chosen factors? (time series, GRS)
 - ▶ Risk premium: is a factor's premium nonzero? (a traded factor: its mean excess return; cross-section: Fama–MacBeth λ)

Time-Series Test: GRS

- N test assets, K traded factors, T periods: $r_{it}^e = \alpha_i + \beta_i' f_t + \varepsilon_{it}$; $H_0: \alpha_1 = \dots = \alpha_N = 0$
 - ▶ $GRS = \frac{T - N - K}{N} \frac{\hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \bar{f}' \hat{\Omega}^{-1} \bar{f}} \sim F(N, T - N - K)$ [Gibbons, Ross & Shanken \(1989\)](#)
 - ▶ $\hat{\Sigma}$: residual covariance, $\hat{\Omega}$: factor covariance (both divided by T); the denominator is one plus the squared Sharpe ratio of the factors
- The F distribution is exact only if the errors, conditional on the factors, are i.i.d. with the Normal distribution; crypto returns have $\hat{\alpha}_H$ near 3 and volatility clustering, so the residuals hardly are
 - ▶ Robust alternatives: a GMM Wald test with HAC covariance, or a wild bootstrap under H_0 : $r_t^{e*} = \hat{\beta}' f_t + \hat{\varepsilon}_t \eta_t$, one Rademacher η_t for all assets (keeps the cross-correlation)
- $\hat{\Sigma}$ must be invertible and precise: N much smaller than T ; weekly crypto data since 2018 give $T \approx 450$
- GRS tests the joint pricing restriction, not a factor's premium: a zero-premium factor need not cause a rejection, and a priced factor in an incomplete model can fail GRS

Cross-Section: Fama–MacBeth, Shanken and Weak Factors

- Two passes: $\hat{\beta}_i$ from time series; then each week $r_{it}^e = \gamma_0 + \hat{\beta}_i' \lambda_t + e_{it}$; $\hat{\lambda} = \bar{\lambda}_t$, $SE_{FM} = sd(\hat{\lambda}_t)/\sqrt{T}$
 - ▶ The FM standard error ignores that $\hat{\beta}_i$ is estimated (errors in variables): too small
 - ▶ [Shanken \(1992\)](#), for $\hat{\theta} = (\hat{\gamma}_0, \hat{\lambda}')'$ and $X = [1, \hat{\beta}]$: $\widehat{Var}(\hat{\theta}) = \frac{1}{T} [(X'X)^{-1} X' \hat{\Sigma} X (X'X)^{-1} (1 + \hat{\lambda}' \hat{\Sigma}_f^{-1} \hat{\lambda}) + \hat{\Sigma}_f^*]$,
 $\hat{\Sigma}_f^* = \begin{pmatrix} 0 & 0 \\ 0 & \hat{\Sigma}_f \end{pmatrix}$; $\widehat{Var}(\hat{\lambda})$ is its lower $K \times K$ block
 - ▶ Assumptions: residuals serially uncorrelated and homoskedastic conditional on the factors, $\hat{\Sigma}$ their covariance; the residual part is multiplied by $1 + \hat{\lambda}' \hat{\Sigma}_f^{-1} \hat{\lambda}$, the factor part $\hat{\Sigma}_f^*$ is added once
- Weak or useless factors (small spread of β , e.g. attention proxies): λ is weakly identified and FM t -statistics over-reject
 - ▶ Use identification-robust tests of risk premia [Kleibergen \(2009\)](#)
- Multiple testing: dozens of crypto “anomalies” have been tried; a new factor should clear $|t| > 3$ [Harvey, Liu & Zhu \(2016\)](#), or control the false discovery rate

Data-Quality Threats to Identification

- **Wash trading:** fake volume on unregulated exchanges inflates volume- and liquidity-based factors [Cong, Li, Tang & Yang \(2023\)](#)
 - ▶ Filter the exchanges and volumes as in the cited paper before building any factor
- **Which price?** Large, recurrent gaps between exchanges, much larger across countries than within them [Makarov & Schoar \(2020\)](#)
- **Survivorship:** today's large coins are the winners; dead coins are missing from most samples [Będowska-Sójka, Wójcik & Pele \(2026\)](#)
- **Remedy:** fix the investable universe ex ante (exchange list, volume filter), keep delisted coins, report results by sub-universe

Worked Result: Eight Coins and the Crypto Market Factor

- Test assets: Bitcoin, Ethereum, XRP, BNB, Cardano, Dogecoin, Litecoin, Chainlink; weekly excess returns over the 4-week Treasury bill (FRED DTB4WK), $T = 450$
 - ▶ Factor: excess return of the five-coin market-value index of Section 3; CSIZE and CMOM cannot be built as in Liu, Tsyvinski & Wu (2022): they need long-short portfolios over a broad cross-section with circulating market values, public here for five coins only
- GRS = 2.16: $p = 0.029$ from $F(8, 450 - 9)$, $p = 0.018$ from the wild bootstrap: zero alphas rejected at 5%
 - ▶ Alphas: Dogecoin +99% a year ($t = 1.23$), BNB +51% ($t = 1.97$), Chainlink +48%; no single $|t|$ above 1.97
- Fama-MacBeth: $\hat{\lambda} = 163\%$ a year, SE 147 (FM) and 156 (Shanken); the market factor itself earned 39% ($t = 1.81$)
 - ▶ Betas range only from 0.90 to 1.40: eight coins cannot identify a premium (a weak cross-section)
- Reading: the rejection most likely reflects survivors (coins chosen because they are large today), not a new factor

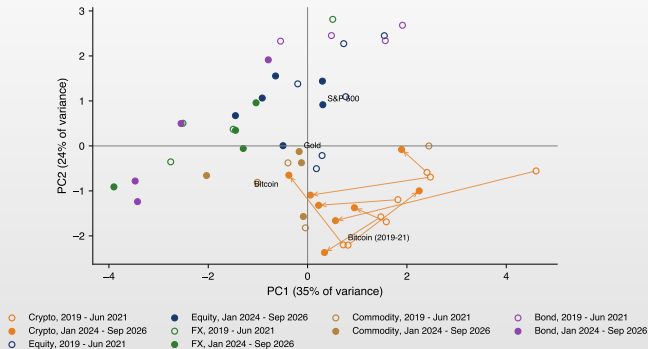
A Separate Asset Class? The Evidence So Far

- [Pele et al. \(2023\)](#) compare cryptos with stocks, exchange rates, commodities, bonds and real-estate indices
 - ▶ Dimension reduction: most of the variation is explained by **tail**, **memory** and **moment** factors of log returns
 - ▶ Classification methods place cryptos in a separate class, mainly because of the tail factor
 - ▶ The tail factor is a feature-space factor (76% of the variance), loading on quantiles, conditional tail expectations, variance and the α -stable tail and scale parameters (Section 3.2): it mixes scale with shape and is not a difference in Hill indices
 - ▶ Over time cryptos became more alike and more distinct from other assets
- Hedge or safe haven? A hedge is uncorrelated or negatively correlated on average; a safe haven on the worst market days [Baur & Lucey \(2010\)](#)
- Early evidence: Bitcoin a diversifier, a hedge only in some periods [Bouri et al. \(2017\)](#); we test it with post-ETF data in Seminar 16, Part C

A Small Replication: Features and Method

- 26 assets: 8 cryptos, 6 equity indices (including BET), 4 exchange rates, 4 commodities, 4 bond ETFs; two windows
 - ▶ W1: January 2019 – June 2021; W2: 11 January 2024 – 18 September 2026 (after the spot ETFs)
- Eight features of daily log returns, each asset on its own calendar
 - ▶ Moments: $\ln \sigma_a$, skewness, $\ln(1 + K)$; memory: $\hat{\rho}_1(r)$, $\hat{\rho}_1(r^2)$
 - ▶ Tails: Hill indices of losses and gains (5% largest); maximum drawdown
- **Principal component analysis (PCA)** of the standardised features (Chapter 3): directions of maximum variance
 - ▶ Distance between class centroids in the full 8-dimensional standardised space

Cryptos Among Other Asset Classes



- First two principal components; hollow markers: 2019 – June 2021, filled: January 2024 – September 2026; arrows: each crypto from W1 to W2

Reading the Asset-Class Map

- PC1 and PC2 explain 35% and 24% of the variance; PC1 mainly measures volatility, kurtosis, drawdown depth and tail heaviness
 - ▶ Cryptos sit apart in both windows, as in [Pele et al. \(2023\)](#)
- Point estimates move **towards** the other classes after 2024: distance to equities from 3.03 to 2.63, to commodities from 2.26 to 1.86
 - ▶ Bitcoin's volatility fell from 77% to 47%; median crypto Hill index rose from 2.32 to 3.26 (thinner tails; 6 of 8 still below 4)
- Cryptos also became more alike: average distance to the crypto centre from 2.13 to 1.43
- Two windows and 26 assets are a small sample: are these moves larger than sampling noise? Next slide

Is the Move Significant? A Block Bootstrap of the Map

- In each window, draw the same calendar days for all 26 assets (moving blocks of 28 days: four weeks, 20 trading days); each asset keeps its own returns on those days, so cross-asset dependence is preserved; recompute the eight features, the common standardisation and the centroid distances; 500 replications
 - ▶ Statistic: change $\Delta = d_{W2} - d_{W1}$ in the distance between the crypto centroid and each class centroid
- Equities: $\Delta = -0.39$, 95% CI $[-1.04, 0.59]$; commodities: -0.41 $[-1.38, 0.47]$; both contain 0
 - ▶ Currencies: -0.89 $[-1.67, 0.18]$; bonds: -0.62 $[-1.55, 0.92]$
 - ▶ Spread of the cryptos around their centre: -0.70 $[-1.09, 0.31]$
- Reading: all five intervals contain 0: every move is within sampling noise; cryptos remain distinct, and the post-ETF moves towards the other classes are not significant
- Caveat: $\ln(1 + K)$ and $\hat{\rho}_1(r^2)$ have population targets only if $Er^4 < \infty$, doubtful here: they enter as sample descriptors, so the map is descriptive and these intervals are indicative

Recap: Alternative Assets

- ▣ Crypto factors (market, size, momentum): GRS judges the model's alphas, Shanken-corrected Fama–MacBeth the premia; both need data free of survivorship bias
- ▣ Tail, memory and moment features separate cryptos from other asset classes [Pele et al. \(2023\)](#)
- ▣ Since the spot ETFs cryptos are calmer, but their move towards equities is not significant: still a distinct class
- ▣ Positive correlation with shares since 2020: a diversifier at best, not a safe haven by default

Case Study: Are Cryptocurrencies Priced Like Stocks, Currencies or Gold?

- [Liu & Tsyvinski \(2021\)](#), *Review of Financial Studies*: a landmark study that prices cryptocurrency returns with standard asset-pricing tools
 - ▶ Question: is the risk–return trade-off of Bitcoin, Ripple (XRP) and Ethereum that of a known asset class, or is it driven by factors specific to crypto markets?
- Three popular narratives, each a testable exposure
 - ▶ A stake in blockchain technology, like a share: loadings on stock-market factors
 - ▶ A medium of exchange, like a currency: loadings on exchange rates
 - ▶ A store of value, like gold: loadings on precious metals
- Link to this chapter: the stylised facts of Section 2, the crypto factors and the GRS test of Section 7, the hedge and safe-haven tests [Baur & Lucey \(2010\)](#)
 - ▶ The predictors found here (momentum, attention) became the cross-sectional factors CMKT, CSIZE, CMOM of [Liu, Tsyvinski & Wu \(2022\)](#)
- Numbers on the next slides: tables of the working-paper version [Liu & Tsyvinski \(2018\)](#)

Case Study: Data and Method as Published

- Daily prices: Bitcoin 1 January 2011 – 31 May 2018, Ripple from 4 August 2013, Ethereum from 7 August 2015 (Section 2)
 - ▶ Daily, weekly (Sunday to Sunday) and monthly returns; benchmarks: US stocks, five major currencies against the dollar, gold, silver, platinum, macroeconomic growth rates, 155 anomaly portfolios
- Exposures: monthly regressions $r_t = \alpha + \beta' f_t + \varepsilon_t$ with the CAPM, three-, four-, five- and six-factor stock models [Fama & French \(2015\)](#) (Tables 5–7), currencies (Table 8), metals (Table 11), macro factors (Table 13)
 - ▶ Inference: every t -statistic twice, with the usual and with bootstrapped standard errors (Section 2)
- Crypto-specific predictors: $r_{t+h} = a + b x_t + e_{t+h}$, $h = 1, \dots, 7$ (Tables 14–31)
 - ▶ x_t : the coin's own return (time-series momentum [Moskowitz, Ooi & Pedersen \(2012\)](#)); Google searches for the coin's name minus their average over the previous four weeks, standardised (attention [Da, Engelberg & Gao \(2011\)](#)); Twitter counts; price to wallet users; realised volatility
 - ▶ Quintile sorts: weeks ranked by this week's x_t , mean return of each quintile over the next 1–4 weeks (Tables 15, 20); cut-offs from the first two years only (Tables 17, 22)

Case Study: Key Results

Result	Numbers in the paper	Reading
Weekly Bitcoin (Table 1)	mean 3.79%, standard deviation 16.64%, Sharpe ratio 0.23 (stocks 0.13), skewness 1.76 (stocks -0.38)	far riskier than stocks, yet paid more per unit of risk, with a right tail
Daily extremes (Table 1, Panel B)	$P(r < -20\%) = 0.48\%$, $P(r > 20\%) = 0.95\%$	"miracles" twice as frequent as "disasters"
Stock factors (Table 5)	CAPM: $\alpha = 18.91\%$ a month ($t = 2.42$), $\beta = 3.34$ ($t = 1.45$); five factors: $\alpha = 16.72\%$ ($t = 2.07$); R^2 0.02–0.05	stock factors explain at most 5% of the variance
Currencies (Table 8)	no significant loading on any of the five currencies; α 23.41–24.27% a month; $R^2 \leq 0.01$	not priced like a currency
Momentum (Tables 14, 15)	weekly $b = 0.19$ ($t = 3.73$; bootstrap 2.17): one standard deviation adds 3.16% next week; top quintile 11.22% a week, bottom 2.60%	weekly trends persist for up to four weeks
Attention (Tables 19, 20)	one standard deviation of Google searches: +1.84% ($t = 2.17$) after one week, +2.30% ($t = 2.81$) after two; quintiles 11.20% versus 1.07%	investor attention forecasts returns

- Daily momentum: $b = 0.06$, $t = 2.96$ with the usual errors but 1.22 with the bootstrap (Table 14, Panel A): only the weekly effect survives both

Case Study: Replicate, Limits, Extension

- **Replicate** with the course data: daily closes of Bitcoin, XRP and Ether; stock factors and the risk-free rate from the [Kenneth French Data Library](#); exchange rates from FRED
 - ▶ On the published sample (to 31 May 2018): rebuild Table 1, the monthly regressions of Table 5 and the weekly momentum of Tables 14–15, with both kinds of t -statistics
 - ▶ Data limit: search and Twitter counts are not in the course data, so Tables 19–29 cannot be rebuilt here
- **Limits** stated by the authors
 - ▶ Short samples: Ripple starts in 2013, Ethereum in 2015 (Tables 2–3); quintile cut-offs from the full sample use future data, hence the reruns with cut-offs from the first two years (Tables 17, 22)
 - ▶ For Ethereum momentum is weaker than for Bitcoin and Ripple (Table 18)
- **Open extension:** out of sample, June 2018 – September 2026, is weekly momentum still there, and did the stock beta rise after the spot ETFs of January 2024?
 - ▶ Same regressions, a break test at an unknown date, and a family of tests reported in full (Seminar 16, C3)

An Open Question

Did the spot ETFs integrate Bitcoin with equities?

- US spot Bitcoin ETFs have traded since 11 January 2024: did they change how Bitcoin co-moves with equities in the mean, in the tails and in price discovery?
 - ▶ A live question: [Lee & Lee \(2025\)](#) study how Bitcoin's links with financial markets shifted after the approval; [Kang & Lee \(2026\)](#) title their answer “institutionalization without integration”
- Formal version: $H_0: \rho_{\text{post}} = \rho_{\text{pre}}$ against $H_1: \rho_{\text{post}} > \rho_{\text{pre}}$; the same for the lower-tail dependence λ_L of Chapter 6
 - ▶ Falsified if the break date chosen by the data lies far from January 2024, or if placebo dates give changes as large
- Identification: the launch is an event, not an experiment
 - ▶ It overlaps with the rate cycle, the halving of the mining reward in April 2024 and ETF flows that themselves respond to prices

The Discovery Loop, Step by Step (1/2)

- **Literature:** map what is known, then verify every reference
 - ▶ “List 2023-2026 papers that test for a change in Bitcoin-equity co-movement at the US spot ETF approval: DOI, sample, method, sign of the effect.”
 - ▶ Resolve each DOI on [Crossref](#), open the paper, find the number it is cited for; search with Semantic Scholar, Elicit or Google Scholar
- **Hypotheses:** competing mechanisms, each with an implication the others do not share
 - ▶ “Give three competing mechanisms for the post-2020 Bitcoin-S&P 500 correlation and, for each, one testable implication that separates it from the others.”
 - ▶ E.g. ETF flows predict an effect concentrated in US trading hours; a common monetary factor predicts co-movement with rate surprises already before 2024
- **Data and code:** on the course data, replicating a known number first
 - ▶ “Read data/market/BTC-USD.CC.csv and GSPC.INDX.csv, join the prices on common days, then compute log returns and their 2020-2023 correlation.”
 - ▶ Replicate first: this chapter reports 0.39 (2020–2023) and 0.38 (since the ETFs); no new result before the code hits both

The Discovery Loop, Step by Step (2/2)

- **Robustness**, fixed before the post-ETF results are seen
 - ▶ “Draft a pre-registered robustness plan: break-date scan, placebo launch dates, weekly returns, tail dependence, block bootstrap; fix every tuning choice now.”
 - ▶ The plan fixes windows, block length and the family of tests: the number of specifications is known and the FWER can be controlled
- **Critique**: ask the assistant to attack the result
 - ▶ “Act as a referee: give the three strongest reasons this change in correlation could be spurious, and the test that would settle each.”
 - ▶ Expect: non-synchronous closes, correlation that rises mechanically with volatility Forbes & Rigobon (2002), confounding events
- **Report**: effect size with a CI, every specification tried, what failed
 - ▶ The prompts that shaped the design go into AI_USE.md

What the Researcher Must Check

- **Calendars:** returns computed separately and then joined by date pair Bitcoin's Sunday-to-Monday move with the S&P 500's Friday-to-Monday move; Bitcoin's Friday-to-Sunday moves are lost
 - ▶ Test: join prices first; with no missing Bitcoin days, the number of common days equals the S&P 500 trading days in the sample; otherwise list the missing dates
- **Non-synchronous closes:** the Bitcoin daily close and the New York close are hours apart, which biases the daily correlation towards zero
 - ▶ Test: weekly returns and the lead-lag correlation $\text{corr}(r_{t+1}^{\text{BTC}}, r_t^{\text{SPX}})$
- **Volatility bias:** Pearson correlation rises when volatility rises, without any change in the link [Forbes & Rigobon \(2002\)](#)
 - ▶ Test: the volatility-adjusted correlation, or the contrast within volatility regimes
- **Survivorship and supply:** today's large coins only, total instead of circulating supply
 - ▶ Test: the universe as it was on each date; circulating supply, as in the CRIX rule
- **Invented references and numbers**
 - ▶ Test: DOI resolves on Crossref with the same title; every number is regenerated by code in the repository

Mini-Case: Did the ETFs Break the Correlation?

- A typical assistant proposal: “the spot ETFs integrated Bitcoin with equities”; we test $H_0: \rho_{\text{post}} - \rho_{\text{pre}} = 0$ on Bitcoin and the S&P 500
- Pre/post contrast: $\hat{\rho} = 0.39$ in 2020–2023 ($n = 1\,006$) and 0.38 since 11 January 2024 ($n = 674$)
 - ▶ Difference -0.01 , 95% CI $[-0.12, 0.13]$ from a circular block bootstrap (blocks of 20 days, 5 000 replications)
- Let the data choose the date: $z(\tau) = [\text{atanh } \hat{\rho}_{\text{after}} - \text{atanh } \hat{\rho}_{\text{before}}] / \sqrt{2/(W-3)}$, $W = 500$ common days on each side of τ
 - ▶ Largest break on 5 Feb 2020 ($z = 7.26$); at the ETF date $z = -1.48$, and 82% of candidate dates give a larger $|z|$
- Interpretation: no evidence that the ETFs moved the mean correlation; the integration came in early 2020, as the chapter shows; tails and price discovery remain open

Your Turn: a Project Seed

- **Question:** did institutionalisation change Bitcoin's **tail** dependence and **price discovery** with US equities?
- **Replicate first:** $\hat{\rho} = 0.39$ and 0.38 from this chapter, then one headline estimate of [Lee & Lee \(2025\)](#) or [Kang & Lee \(2026\)](#) on their own sample
 - ▶ Extend: lower-tail dependence from copulas (Chapter 6), weekly returns, IBIT–Bitcoin lead–lag in US trading hours
- **Deliverables:** a pre-registered plan committed before the results, a notebook that regenerates every number, effect sizes with CIs and the full list of specifications
- **AI log:** prompts that shaped the design in `AI_USE.md`; each wrong claim (an invented DOI, returns on mismatched calendars) and the test that caught it in `AI_ERRORS.md` ([templates](#))

Common Pitfalls

- ▣ Reporting sample kurtosis when $\hat{\alpha}_H < 4$: it is not a consistent estimator, and its bootstrap CI is invalid
- ▣ Testing a threshold or a break date with χ^2 critical values: the nuisance parameter is not identified under H_0
- ▣ Comparing daily closes recorded at different times and concluding that an ETF tracks badly
- ▣ Reading TVL growth as adoption when it is mostly the Ether price
- ▣ Treating a stablecoin as risk-free cash: the peg is only as good as redemption and reserves
- ▣ Total instead of circulating supply in a market-value index; survivorship: studying only today's large coins ignores the many that failed [Będowska-Sójka, Wójcik & Pele \(2026\)](#)

Coming Next

- Chapter 17: bubbles and crashes: explosive-root tests and log-periodic power laws, with crypto booms as a main case
- Chapter 18: systemic risk and networks: spillovers between crypto, banks and equity markets
- Earlier chapters revisited: VaR and ES for crypto (Chapter 7; high-frequency entropy helps forecast Bitcoin's VaR [Pele & Mazurencu-Marinescu-Pele \(2019b\)](#)), liquidity and order books on crypto exchanges (Chapter 10) [Agakishiev, Härdle & Pele \(2025\)](#)
- Further reading: [Härdle, Harvey & Reule \(2020\)](#), [Franke, Härdle & Hafner \(2019\)](#), [Schär \(2021\)](#)

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Are Cryptos becoming alternative Assets?
- ▣  Introduction to Blockchain and Cryptocurrencies
- ▣  Comparison of Cryptocurrency Indices
- ▣  Cryptocurrency as an Asset Class
- ▣  Smart Contracts
- ▣  Stabilising Onchain Stablecoin Risks
- ▣  Risk Premia in the Bitcoin Market

Code and data

- ▣ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_16

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